LASER CUTTING & ENGRAVING

Focused light as a cutting tool — beam physics · laser types · machine anatomy · parameters · materials · safety · process comparison · applications

10^9 W/cm²	beam focused to a tiny spot — power density
0.2–0.4 mm	kerf — the width of material removed
HAZ very small	heat-affected zone — crisp, clean cut edges
±0.05–0.1 mm	repeatable tolerance on finished parts
~25 mm	steel cut by a high-power fiber laser

THE BASICS

WHAT IS LASER CUTTING?

Light Amplification by Stimulated Emission of Radiation

A laser cutter uses a **focused, high-power beam of light** to melt, burn or vaporise material. Because the energy is concentrated to a tiny spot, the material at the point of contact is heated to thousands of degrees in milliseconds and **ablated away** — leaving a narrow, clean slot.

The three removal modes

- **Melt** — beam melts a kerf; a gas jet blows the molten pool out (metals)
- **Burn / vaporise** — material combusts or turns straight to vapour (wood, acrylic)
- **Combined** — most sheet cutting uses a melt + gas-jet combination

An **assist gas jet** (CO■, N■ or compressed air) is aimed at the cut point under pressure. It does two jobs at once: it **blows away** molten or vaporised material, and it **protects** the lens and cools the edge. The gas also controls whether the edge oxidises: nitrogen gives a clean, oxide-free bright edge on stainless, while oxygen accelerates cutting of mild steel.

Why it matters: laser cutting is **non-contact** — nothing touches the work, so there is no tool wear, no mechanical force and no distortion of the sheet. A single tool can cut a filigree pattern one hour and engrave a photograph the next, purely by changing software.

Non-contact

- No cutting edge to dull — beam never wears out
- No cutting force — thin, fragile parts cut cleanly
- Unlimited shapes from vector paths — kerf is the only offset

THE MACHINE

HOW A LASER CUTTER WORKS

From cavity to cut — the optical + motion chain

1. The cavity / resonator — the heart. Two mirrors face each other with a gain medium (gas, crystal or fibre) between them. Energy (electrical or optical) 'pumps' the medium, and light bounces back and forth, gaining intensity at each pass until a powerful coherent beam emerges through the partially-reflective **output coupler**.

2. Beam delivery — in a CO■ machine the beam travels along folded **mirror** arms mounted on the gantry (mirror tiles keep it aligned). In a fiber laser the beam runs down a flexible **optical fibre**, so no mirrors are needed.

3. The focusing lens — a convex lens (or focussing head) squeezes the collimated beam down to a spot of a few tenths of a millimetre. The **focal length** sets the depth of focus: a short lens gives a small spot for fine engraving, a longer lens gives a deeper focus for thick stock.

4. The nozzle + assist gas — the focused beam exits through a nozzle; high-pressure gas (O■ / N■ / air) surrounds it to purge and cool the cut.

5. The CNC X–Y gantry — stepper or servo motors drive the head across the bed on rails in X and Y (and sometimes a Z axis for focus height), following G-code generated by CAM software.

6. Fume extraction — a fan under the honeycomb bed and a duct pull smoke, vapour and particulates to a filter or outside; an interlock kills the beam if the lid opens.

THE SOURCES

LASER TYPES — WHAT & WHY

Wavelength decides what the beam can cut or mark

The **wavelength** determines which materials absorb the light. Organics and acrylics absorb the 10.6 µm of CO■; metals barely do (they reflect it). Metals absorb the 1.06 µm of fiber/ytterbium. This is the single most important choice when buying a machine.

CO■ (gas) — 10.6 µm, 40–150 W
The workshop/sign-shop classic. Cuts wood, acrylic (including clear), MDF, leather, fabric, paper, rubber. Great clean edges on acrylic. Cannot cut bare metal — the beam reflects.

Fiber / ytterbium — 1.06 µm, 500 W–10 kW+
The industrial metal cutter. Cuts and deeply marks steel, stainless, aluminium, brass, copper, titanium. Small spot, very fast, low running cost, no gas tube to replace.

Nd:YAG — 1.064 µm pulsed
Older solid-state crystal laser; good for spot welding and fine metal marking where higher pulse energy is useful. Largely superseded by fiber lasers.

Diode — 405–455 nm blue / 450 nm
Hobby diode lasers mark dark woods, paper, leather and anodised aluminium. Cheap and air-cooled, but low power (usually under 20 W) — cannot cut clear materials or bare metal.

Type	Wavelength	Power	Best for
CO■ gas	10.6 µm	40–150 W	Wood, acrylic, non-metal
Fiber / Yb	1.06 µm	0.5–10 kW	Metal cutting & marking
Nd:YAG	1.064 µm	pulsed	Fine metal marking / welding
Diode	405–455 nm	1.5–20 W	Hobby wood / leather / mark

WHAT IT DOES

CUT vs ENGRAVE vs MARK

Three jobs, one machine — plus raster vs vector

Job	What happens	Depth
Cutting	Beam passes fully through the sheet, separating two pieces	Thickness of sheet
Engraving	Beam removes a shallow layer to create a recessed image or text	0.1–2 mm
Marking	Beam alters the surface colour or oxide layer only — no real depth	surface / µm

Cutting is full penetration — the kerf opens right through. **Engraving** is partial-depth removal, sculpting a groove so light-and-shadow shows the design. **Marking** doesn't remove material: it changes the surface — annealing metal to form dark oxide, or foaming/charring plastic for colour contrast. Marking is fastest and leaves the surface flush.

Raster vs vector modes

- **Raster** — the beam scans back and forth line by line like a printer, power on/off per pixel. Used for engraving photos, shading and tonal images; slower but gives greyscale depth.
- **Vector** — the beam follows continuous outlines/curves like a pen, cutting lines fast. Used for clean straight cuts, outlines and text; much faster per distance than raster.
- Most jobs mix both: vector-cut the outline, raster-fill the interior shading.

Kerf compensation: because the beam removes a real width, vector paths are offset inward by half the kerf so the finished part meets the drawn size exactly. Software (LightBurn, LaserGRBL) does this automatically.

TUNING THE CUT

PARAMETERS — POWER, SPEED & MORE

Every variable changes edge quality & speed

Five core dials trade off speed against edge quality. All are tuned with a **test grid** — a matrix of small squares cut at stepped power/speed combinations, so the best combo is read off directly.

- **Laser power (W)** — the raw output of the tube/fiber; larger W = thicker stock and faster cuts, but more heat and char risk.
- **Speed (mm/s or mm/min)** — how fast the head moves. Too fast = doesn't cut through; too slow = charred, melted edges (especially wood/acrylic).
- **Power %** — the commanded fraction of maximum. Cutting a 6 mm board might use 80–100%; engraving uses 10–40% so it scorches, not severs.
- **Frequency / Hz (pulse rate)** — for CO■, pulses-per-second. High frequency ≈ continuous beam (smooth cut); low frequency = spaced dots (useful for engraved shading).
- **Focal position** — where the focal point sits relative to the surface. At the surface = best fine cut; slightly into/below the material = better thick cuts; raise it for flame-polished engraving.
- **Air-assist pressure** — more pressure clears molten material and cuts faster but can blow the sheet; less pressure lets flames burn cleaner on wood (less char).
- **Pass count** — number of times the beam repeats the path. Multiple light passes cut thick material with less heat than one heavy pass.

Cut vs engrave settings

CUT: high power (80–100%), moderate speed, high frequency, focus at/near surface, medium-high air, 1–2 passes. ENGRAVE: low power (10–40%), fast speed, lower frequency for shading, focus raised for a wider softer mark, low air to preserve fine detail.

Test-first: on a new material always run a small power/speed matrix, check cut-through and char, then scale up. Document settings in a material library to reuse.

MATERIALS

WHAT CUTS — & WHAT NEVER TO CUT

Absorption decides; chemistry decides safety

CO■ lasers cut a wide range of **non-metallic** materials cleanly. Fiber lasers are the tool for **metal**. Read the material chemistry before cutting — some plastics release deadly fumes.

CUTS WELL (CO■)

- **Wood** — plywood, MDF, balsa, hardwoods; clean edges, light char
- **Acrylic / PMMA** — the star material: flame-polished bright edges, great for signs
- **Cardboard & paper** — fast, cheap prototypes and packaging
- **Leather & suede** — engraving, inlays, bespoke goods
- **Fabric** — cotton, felt, polyester; laser-seals edges so they don't fray
- **Cork, rubber, EVA foam** — gaskets, stamps, mats
- **Glass & stone** — engravable (marks, doesn't cut)

METALS (FIBER)

- Steel, stainless, aluminium, brass, copper, titanium — sheet metal cutting and deep marking
- CO■ mostly bounces off bare metal — use paint/coating + mark, or switch to fiber

NEVER CUT — TOXIC / DANGEROUS

- **PVC / vinyl / vinyl chloride** — releases **chlorine gas** (corrosive, toxic) + hydrochloric acid that destroys the machine
- **ABS** — melts, catches fire and emits cyanide fumes; smoky and messy
- **Coated / galvanised / painted metals** — fumes from the coating are toxic
- **PTFE / Teflon**, **polycarbonate** (clouds & burns), **fiberglass / carbon-fibre** (irritant fibres + resin gas), **HDPE/bottle plastic** (melts & catches fire), **treated or pressure-treated wood** — all unsafe

Rule of thumb: if you can't confirm what the material is made of, don't cut it. When in doubt, test a tiny corner and watch for smoke colour and odour.

WHY CHOOSE IT

ADVANTAGES OF LASER CUTTING

Where the focused beam beats mechanical tools

- **Tiny kerf** — typically 0.2–0.4 mm, so little material is wasted and intricate, tightly-packed parts fit on a sheet; detail finer than any saw or router.
- **High precision & detail** — repeatable tolerance of ±0.05–0.1 mm; cuts filigree, micro-text and fine serif typography cleanly.
- **Non-contact, no tool wear** — the beam never dulls and nothing physically pushes on the work; no cutting forces, no deflection, no tool-change downtime.
- **Very small heat-affected zone (HAZ)** — energy is concentrated so tightly that surrounding material barely heats; edges stay crisp with minimal distortion, discolouration or burr.
- **Fast on thin material** — a single pass can sever sheet in seconds; hundreds of parts per hour; no setup tooling beyond the digital file.
- **Fine engraving** — raster engraving reproduces photographs, textures and gradients in wood, acrylic and coated metal that no mechanical engraver matches.
- **No secondary finishing** — on many materials the cut edge is the finished edge (flame-polished acrylic, sealed fabric).
- **Digital / flexible** — change the design in software and re-cut instantly; perfect for one-offs, personalisation and small runs.

KNOW THE LIMITS

LIMITATIONS OF LASER CUTTING

What the beam struggles with

- **Thick metal is limited** — even a 10 kW fiber struggles beyond ~25–30 mm steel at practical cost; plasma or waterjet wins on thick plate. CO■ can't cut bare metal at all.
- **Heat-affected-zone edges** — the cut edge of metals still shows a thin HAZ and a recast layer; aerospace and medical parts often need a final clean skim.
- **Fumes & ventilation are mandatory** — vaporised material and smoke must be extracted and filtered; the machine is unusable without it.
- **Reflective metals** — copper, brass and polished aluminium reflect the beam back, damaging the optics (fiber handles them better than CO■).
- **Fire risk with some woods** — pine knots and resinous species can ignite; any operator must watch for flames in the cut.
- **High machine cost** — a capable CO■ or fiber laser is expensive; industrial metal cutters run into six figures; optics need occasional cleaning and gas tubes/Diode sources wear out.
- **Taper & edge squareness** — the converging beam gives a slightly tapered wall on thick stock rather than perfectly vertical faces.
- **No multi-axis complex 3-D** — laser cutters are essentially 2-D sheet tools; true 3-D contours need routing, milling or waterjet with a tilt head.
- **Material-specific** — plastics like PVC are off-limits, and some composites degrade or smoke badly.

STAY SAFE

SAFETY — BEFORE YOU PRESS START

A 150 W beam can blind, burn and start fires

- **Never run unattended** — a laser can set fire in seconds; stay at the machine for every job. A wooden/resinous part that catches fire must be caught early.
- **Fume extraction always on** — run the fan/filter for the whole job and let it clear after; toxic smoke (from banned materials) must never reach your lungs.
- **Fire watch** — keep a fire extinguisher and a full water container within reach; check the bed and material for smouldering before opening the lid.
- **Eye protection for the wavelength** — CO■ at 10.6 µm is absorbed by ordinary glass/plastic, so standard goggles work; but blue-diode (455 nm) and fiber (1064 nm) beams pass straight through normal lenses — you need certified laser-safety goggles matched to that wavelength and power class (OD rating).
- **Never cut hazardous materials** — PVC, ABS, coated metals, PTFE and the others listed above; chlorine gas and cyanide fumes are genuinely dangerous.
- **Use the enclosure & interlock** — a Class 1 enclosed machine kills the beam the instant the lid opens; pearl on that cover and the E-stop. Never defeat the interlocks.
- **Vent exhaust outside / filter it** — through a window or a charcoal/HEPA filter system; an enclosed room fumes up fast without it.
- **Laser class awareness** — open hobby lasers are Class 4 (the most hazardous) when running; never look directly into the beam or at specular reflections.
- **Keep the optics clean** — smoke slowly films the lens, dimming the beam and worsening edge quality; clean with isopropyl and lens paper.

COMPARE

LASER vs PLASMA vs WATERJET vs CNC ROUTER

Which cutting technology for which job? Each excels on a different axis.

Feature	Laser cutter	Plasma cutter	Waterjet	CNC router
How it cuts	Focused light ablates material	Electric arc + gas jet melts metal	High-pressure water + abrasive erodes	Spinning cutter mills material
Best materials	Wood, acrylic, non-metal, sheet metal (fiber)	Conductive metals — steel, stainless	Almost anything: metal, stone, glass, composites	Wood, plastic, aluminium, soft metal, foam
Thickness limit	~25 mm steel (fiber); small for CO■	Up to 150 mm+ thick plate	300 mm+ thick, any hard material	Limited by spindle & bit; deep 3-D ok
Kerf (width)	0.2–0.4 mm — very small	1.5–3 mm — wide	0.8–1.2 mm + taper	Depends on bit: 2–10 mm
Edge quality	Excellent, clean, small HAZ	Good but dross & HAZ on edge	Excellent, no HAZ, satin finish	Good; needs deburr; no HAZ
Heat-affected zone	Very small	Large (300°C+ near cut)	None (cold cutting)	None (mechanical)
Precision / tolerance	±0.05–0.1 mm (best here)	±0.2–1 mm	±0.05–0.1 mm	±0.1–0.5 mm
Speed on thin sheet	Fastest	Fast	Slow	Medium
Operating cost	Electricity + gas + consumable optics	Consumable nozzles/electrodes + power	High — water, abrasive, energy, disposal	Bits + power
Best for	Fine detail, engraving, signs, small runs	Thick structural steel	Thick hard materials, no-heat cut	3-D carving, pockets, woodwork

WHERE IT'S USED

APPLICATIONS

Signs to stencils to gifts

Application	What it delivers	Why laser
Acrylic signage	Illuminated shop signs, letters, logos	Flame-polished edges, crisp lettering, low cost
Wooden models & puzzles	Scale models, 3-D puzzle kits, ornaments	Tiny kerf = tight joints; intricate shapes
PCBs & stencils	Solder-paste stencils, prototype boards	Fine detail and repeatable accuracy
Fabric & textiles	Cut & sealed garment panels, embroidery appliqué	Edges seal so fabric doesn't fray
Small-batch production	Badges, nameplates, gaskets, brackets	No tooling — switch designs by software
Prototypes & jigs	Design iterations, fixtures, one-offs	Fast turnaround from CAD to part
Personalised gifts	Engraved photos, trophies, jewellery tags	Fine raster engraving reproduces detail
Leather goods	Belts, wallets, stamped branding	Clean burnout with a crisp edge
Architectural models	Scale facades, windows, terrain	Precision + fine fenestration detail
Engraved plaques	Awards, memorials, tags	Deep relief and readable micro-text

TERMS

GLOSSARY — 16 KEY TERMS

The vocabulary that pops up on every job

Wavelength — the distance between wave peaks; decides which materials absorb the beam (CO■ 10.6 µm vs fiber 1.06 µm)

Focusing lens — the convex optic that squeezes the collimated beam down to a tiny high-energy spot

HAZ — heat-affected zone — the thin band of material altered by heat next to the cut

Focal length — lens distance to focus; short = fine detail, long = deeper cuts

Vector — following continuous outlines to cut clean lines fast

Marking — changing surface colour/oxide only — no depth removed

G-code — the numeric motion commands the controller follows

Interlock — a safety switch that kills the beam when the lid opens

Cavity / resonator — the two-mirror optic trap where light bounces and amplifies into the beam

Kerf — the width of material removed by the beam (0.2–0.4 mm) — designs offset for it

Assist gas — O■, N■ or air blown at the cut to blow out melt and cool the edge

Raster — line-by-line scanning that engraves shaded/tonal images

Engraving — removing a shallow layer to create a recessed design

Fume extraction — fan + filter/duct that draws away smoke and vapour from the cut

Gantry — the X–Y motion frame that carries the beam head

Pulse frequency — beam pulses-per-second, dictating cut smoothness vs shading

RECAP

ONE-SENTENCE SUMMARY

A focused laser beam melts/vaporises material, a gas jet clears it away, and software-controlled motion steers the beam for cuts, engraving or marking — fast, precise, non-contact and limited by material chemistry, thickness and safety rules.

WORKFLOW

FROM FILE TO FINISHED PART

The laser job in six steps

- **1. Design** — draw vectors (cut lines) and rasters (engraving) in CAD or Illustrator; export to lightBurn/LaserGRBL.
- **2. Set up the bed** — place material flat on honeycomb/slat bed; tape down thin stock; set a clean, level, dust-free surface.
- **3. Focus** — drop the head to the focus tool/auto-focus so the focal point sits right for the material thickness.
- **4. Tune parameters** — load material-library settings; run a small test grid if new to the material.
- **5. Run with a fire watch** — hit start, keep extraction on, watch for flames, never leave it unattended.
- **6. Finish** — remove parts, clean char/ash, break out tabs, and ventilate before inspecting.

The workflow is digital: the same file can be cut once for a one-off gift or a thousand times for production — which is why laser cutting is the go-to for rapid prototyping and small-batch making.

Pro tips: engrave first, then cut (so the cut doesn't shift the part); keep a material-settings library; clean the lens after smoky jobs; compensate kerf for press-fit joints.